Code No: 126AP **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March - 2024
DISTRIBUTED SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Define synchronous and asynchronous distributed systems. [2]
b) Give some reasons why centralized systems are not adequate to modern computing. [3]
c) Explain distributed termination detection. [2]
d) Differentiate between election algorithm and mutual exclusion algorithm. [3]
e) Explain request-reply protocols. [2]
f) Explain datagram communication in UNIX. [3]
g) Explain cache consistency in Andrew file system. [2]
h) Explain name service requirements. [3]
i) Define wait-for-graph? [2]
j) Define distributed deadlock. [3]

PART - B

(50 Marks)

2.a) Define distributed systems. What are the significant issues and challenges of the distributed systems?
b) List and explain various examples of distributed systems. [5+5]

OR

3.a) Explain how resource sharing is done in the web.
b) Explain in detail about various architectural models of distributed systems. [5+5]

4.a) What is multi cast communication in distributed system?
b) What is distributed mutual exclusion in distributed system? Give examples. [5+5]

OR

5. What are the different ways of synchronizing physical clocks? Explain physical clock synchronization algorithm. [10]

6.a) What is meant by inter process communication? How inter process communication is used in the distributed systems?
b) What are design issues for remote method invocation? [5+5]

OR

7.a) Explain Communication between distributed objects.
b) Write down the steps in java RMI and explain it with suitable programs. [5+5]

8.a) Explain in brief about directory and discovery services.
b) Discuss the design and implementation issues of Domain Name System. [5+5]

OR

9.a) Write about the recent advances in the distributed file systems.
b) Describe basic distributed file system and storage systems and their properties. [5+5]

10.a) Explain two phase commit protocols for nested transactions.
b) Distinguish between three phase commit and two phase commit protocol. [5+5]

OR

11.a) What are the different ways to control concurrency in distributed transaction?
b) How is concurrency control implemented in distributed database? [5+5]

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